

Week 2 — OSINT Data Collection & Data Source Mapping

Course: Cyber Threat Intelligence

Theme: Remote PowerShell Attacks via Windows Remote Management (WinRM) — MITRE ATT&CK T1021.006

Students : Aruzhan Abdulina, Kanasheva Darina

1. Assignment Recap

According to the course syllabus (Week 2 — Data Collection Process), the assignment consisted of three parts:

- Perform OSINT data collection using Shodan, VirusTotal, and Maltego.
- Develop a data source mapping for analysis.
- Recommended reading: Michael Bazzell — Open Source Intelligence Techniques.

All three tools were applied in practice around a single theme — MITRE ATT&CK technique T1021.006 (Remote Services: Windows Remote Management), which describes remote PowerShell command execution over the WinRM service (ports 5985/5986). This is a common lateral-movement technique in real-world attacks, including the activity of the ransomware group Storm-0501.

2. Methodology

The work follows the principle described by Michael Bazzell in Open Source Intelligence Techniques: never rely on a single source, but consistently corroborate (triangulate) findings across several independent OSINT tools. This was implemented as follows:

- MITRE ATT&CK — establishes the formal technical framework (which technique is being investigated, which groups use it).
- Shodan — reveals the real-world scale of the attack surface (how many hosts are potentially exposed).
- VirusTotal — checks the reputation of the discovered infrastructure and analyzes a real malware sample tied to the theme.
- Maltego — visualizes infrastructure relationships and independently corroborates the data obtained through VirusTotal.

3. Technical Context: MITRE ATT&CK T1021.006

Before collecting data, the anchor technique for the whole investigation was established — T1021.006 (Remote Services: Windows Remote Management), part of the Lateral Movement tactic. According to the official MITRE ATT&CK page, this technique is used in real-world attacks by several documented groups and tools, including Cobalt Strike, Brute Ratel C4, Chimera, and the group Storm-0501, which used the post-exploitation tool Evil-WinRM.

MITRE | ATT&CK

MatricesTacticsTechniquesDefensesCTIResourcesBenefactorsContributeBlogSearch

ATT&CKcon 7.0 in-person tickets are open! Join us October 27-28, 2026 in McLean, VA. Register here for in-person tickets; hotel and location details can be found in the FAQ.

TECHNIQUES

VNCWindows Remote ManagementCloud ServicesDirect Cloud VM ConnectionsReplication Through Removable MediaSoftware Deployment ToolsTaint Shared ContentUse Alternate Authentication MaterialCollectionCommand and Control

Home > Techniques > Enterprise > Remote Services > Windows Remote Management

Remote Services: Windows Remote Management

Other sub-techniques of Remote Services (8)

Adversaries may use Valid Accounts to interact with remote systems using Windows Remote Management (WinRM). The adversary may then perform actions as the logged-on user.

WinRM is the name of both a Windows service and a protocol that allows a user to interact with a remote system (e.g., run an executable, modify the Registry, modify services).^[1] It may be called with the winrm command or by any number of programs such as PowerShell.^[2] WinRM can be used as a method of remotely interacting with Windows Management Instrumentation.^[3]

ID: T1021.006
Sub-technique of: T1021
Tactic: Lateral Movement
Platforms: Windows
Version: 1.2
Created: 11 February 2020
Last Modified: 24 October 2025

Version Permalink

Procedure Examples

ID	Name	Description
S1063	Brute Ratel C4	Brute Ratel C4 can use WinRM for pivoting. ^[4]

MITRE | ATT&CK

MatricesTacticsTechniquesDefensesCTIResourcesBenefactorsContributeBlogSearch

TECHNIQUES

VNCWindows Remote ManagementCloud ServicesDirect Cloud VM ConnectionsReplication Through Removable MediaSoftware Deployment ToolsTaint Shared ContentUse Alternate

G1016	FIN13	FIN13 has leveraged winrm to move laterally within a compromised network via application servers and SQL servers. ^[8]
C0048	Operation MidnightEclipse	During Operation MidnightEclipse, threat actors used WinRM to move laterally in targeted networks. ^[9]
S0692	SILENTRINITY	SILENTRINITY tracks TrustedHosts and can move laterally to these targets via WinRM. ^[10]
C0024	SolarWinds Compromise	During the SolarWinds Compromise, APT29 used WinRM via PowerShell to execute commands and payloads on remote hosts. ^[11]
G1053	Storm-0501	Storm-0501 has utilized the post-exploitation tool known as Evil-WinRM that uses PowerShell over Windows Remote Management (WinRM) for remote code execution. ^[12]
G0027	Threat Group-3390	Threat Group-3390 has used WinRM to enable remote execution. ^[13]
G0102	Wizard Spider	Wizard Spider has used Window Remote Management to move laterally through a victim network. ^[14]

MITRE | ATT&CK

MatricesTacticsTechniquesDefensesCTIResourcesBenefactorsContributeBlogSearch

Storm-0501

Storm-1811StriderSuckflyTA2541TA459TA505TA551TA577TA578TeamPCPTeamTNTTEMP.VelesThe White Company

S1040	Rclone	Storm-0501 has utilized Rclone for data exfiltration. ^[2]	Archive Collected Data: Archive via Utility, Data Transfer Size Limits, Exfiltration Over Alternative Protocol: Exfiltration Over Asymmetric Encrypted Non-C2 Protocol, Exfiltration Over Alternative Protocol: Exfiltration Over Unencrypted Non-C2 Protocol, Exfiltration Over Web Service: Exfiltration to Cloud Storage, File and Directory Discovery
S0057	Tasklist	Storm-0501 discovered running processes through tasklist.exe. ^[2]	Process Discovery, Software Discovery: Security Software Discovery, System Service Discovery

References

1. Avertium. (2022, January 11). An In-Depth Look at Ransomware Gang, Sabbath. Retrieved October 19, 2025.

2. Microsoft Threat Intelligence. (2024, September 26). Storm-0501: Ransomware attacks expanding to hybrid cloud environments. Retrieved October 19, 2025.

3. Microsoft Threat Intelligence. (2025, August 27). Storm-0501's evolving techniques lead to cloud-based ransomware. Retrieved October 19, 2025.

4. Tyler McLellan, Brandon Schondorfer. (2021, November 29). Kitten.gif: Meet the Sabbath Ransomware Affiliate Program, Again. Retrieved October 19, 2025.

MITRE

Contact UsTerms of UsePrivacy PolicyWebsite ChangelogCookie Preferences

© 2015 - 2026, The MITRE Corporation. MITRE ATT&CK and ATT&CK are registered trademarks of The MITRE Corporation.

20°C Mostly cloudy

Поиск

KA3

17:17 24.09.2026

From the References section, the Microsoft Threat Intelligence report “Storm-0501's evolving techniques lead to cloud-based ransomware” (August 2025) was selected. This report is the link between the technical framework (MITRE) and the practical OSINT collection: Storm-0501 uses WinRM to gain access and then deploys a ransomware payload, a sample of which (Embargo family) was later analyzed through VirusTotal (see Section 5).

4. Shodan — Collecting Data on Exposed Infrastructure

Query: port:5985,5986 — the two standard ports of the WinRM service (5985 — HTTP, 5986 — HTTPS), which is the transport that PowerShell Remoting (Enter-PSSession, Invoke-Command) physically runs over. Filtered search required a free Shodan account, since anonymous access to filtered queries (port:) is restricted.

The screenshot shows the Shodan search results page for the query 'port:5985,5986'. The page features a dark header with navigation links (Shodan, Maps, Images, Monitor, Developer, More) and a search bar containing the query. The main content area displays 'TOTAL RESULTS: 1,535,797' and a world map highlighting top countries. Below the map is a table of top countries with their respective result counts. The 'TOP PORTS' section shows 5985 and 5986 as the most common ports. The 'Product Spotlight' section highlights 'Free, Fast IP Lookups for Open Ports and Vulnerabilities using InternetDB'. The 'Not Found' section lists 147,931,151,237 results for 'Contabo GmbH' and 'Cloudbase-init WinRM'. The 'SSL Certificate' section shows a certificate issued by 'Cloudbase-init WinRM'.

Country	Count
Singapore	322,995
China	284,827
United States	268,520
Hong Kong	81,362
Israel	76,359

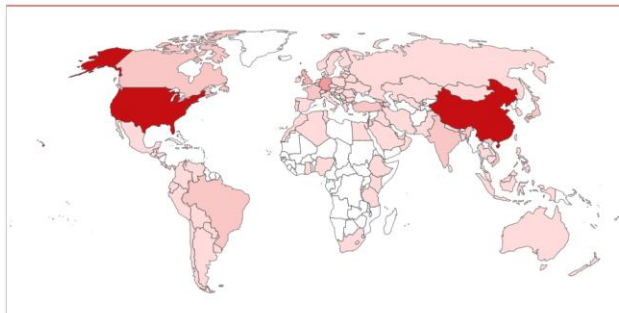
Port	Count
5985	1,040,023
5986	495,775

Shodan Report

port:5985,5986

Total: 1,535,788

// GENERAL



Countries

Singapore	322,994
China	284,825
United States	268,517
Hong Kong	81,362
Israel	76,361

[More...](#)

Ports

5985	1,040,013
5986	495,775

Organization

Incapsula Inc	259,317
ACEVILLE PTELTD.	85,045
Internet Rimon	63,458
Tencent cloud computing (Beijing) ...	52,525
Aliyun Computing Co., LTD	40,430

[More...](#)

Vulnerabilities

HTTP.sys Denial of Service	16
HTTP.sys Remote Code Execution	16
Heartbleed	7
CVE-2018-18441	5
CVE-2021-41277	4

[More...](#)

Products

WinRM	839,706
nginx	17,308
VNC	9,514
OpenSSH	3,164
Socks4A	2,629

[More...](#)

Tags

cdn	277,254
self-signed	147,799
cloud	145,417
eoel-os	8,472
proxy	5,125

[More...](#)

Operating Systems

Windows Server 2022 (build 10.0...	248,123
Windows Server 2012 R2 (build 6...	229,880
Windows Server 2019 (version 18...	168,308
Windows Server 2016 (version 16...	108,461
Windows (build 10.0.26100)	49,781

[More...](#)

// HTTP INSIGHTS

Website Titles	
Not Found	848,814
阿里云 Web应用防火墙	22,382
404 Not Found	11,328
Welcome to OpenResty!	10,008
Welcome to nginx!	6,208
More...	

Web Technologies	
Microsoft HTTPAPI	847,926
Imperva	208,182
Nginx	29,034
OpenResty	12,618
jQuery	2,706
More...	

Protocol Versions	
http/1.1	10,227
h2	16
http/1.0	4

Findings:

- A total of 1,535,423 hosts worldwide were found with WinRM ports open.
- Top countries by number of hosts found: Singapore, China, United States, Hong Kong, Israel.
- Port breakdown: 5985 (HTTP, unencrypted) — 1,039,785 hosts; 5986 (HTTPS) — 495,638 hosts, indicating that the unencrypted configuration of WinRM predominates.
- Example host: 154.57.211.62 (Cogent Communications, Pakistan, Lahore) — the service banner discloses OS Windows Server 2016 (build 10.0.14393), NetBIOS domain, and target name before any authentication attempt, which is itself useful for attacker reconnaissance.

Important methodological note: the discovered host is an example of exposed attack surface, not a confirmed compromised or malicious node. Exposure does not equal compromise — this host merely illustrates the kind of infrastructure that is potentially vulnerable to technique T1021.006.

5. VirusTotal — Reputation Check and Malware Sample Analysis

5.1. Locating a Sample via MalwareBazaar

A direct text search by name (“Storm-0501”, “Embargo ransomware”) in VirusTotal returned no results, since VT’s text search queries community comments rather than the malware family name. The sample was therefore located through a dedicated IOC repository — MalwareBazaar (abuse.ch) — where the Embargo tag lists confirmed samples of this ransomware family (the payload that Storm-0501 and similar groups deploy after lateral movement).

154.57.211.62

No security vendor flagged this IP address as malicious

Community Score: 0 / 91

154.57.211.62 (154.57.208.0/21)
AS 135407 (Trans World Enterprise Services (Private) Limited)

PK Last Analysis Date: 8 months ago

DETECTION DETAILS RELATIONS COMMUNITY

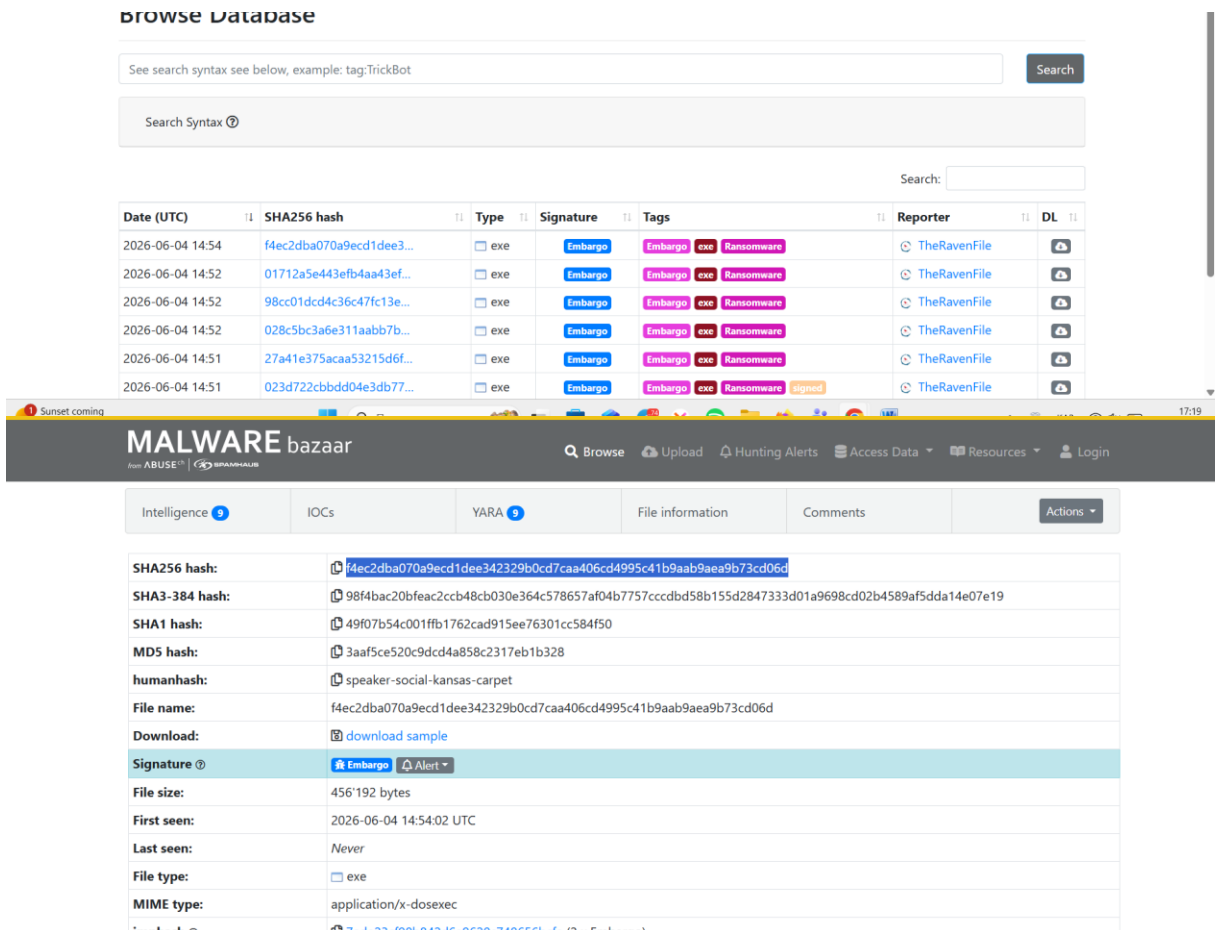
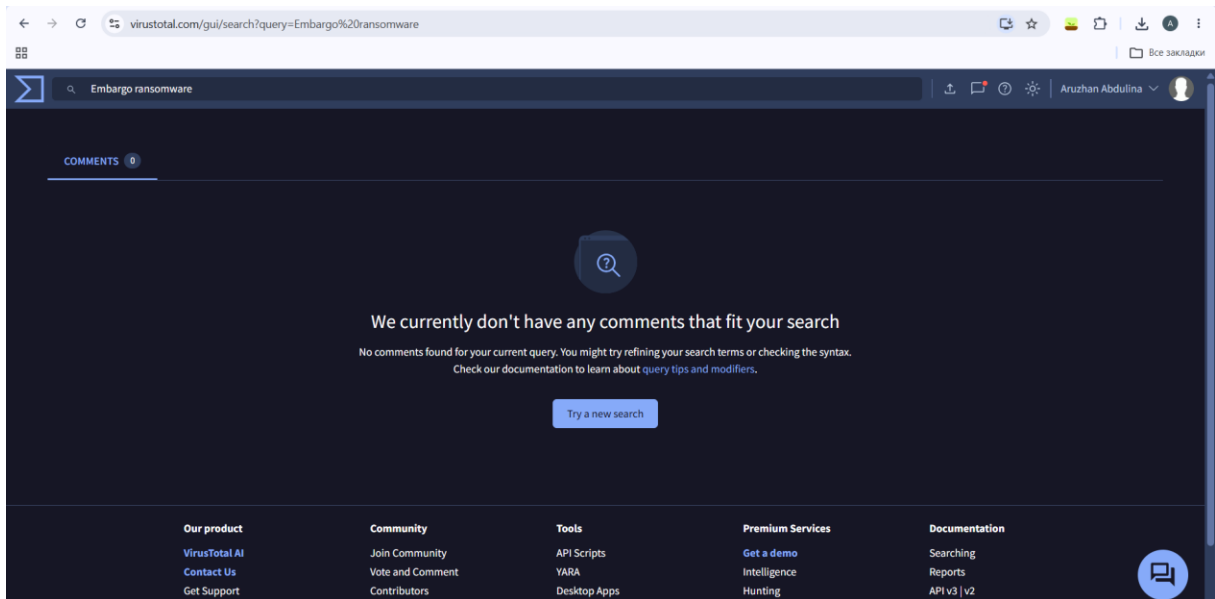
Basic Properties

Network	154.57.208.0/21
Autonomous System Number	135407
Autonomous System Label	Trans World Enterprise Services (Private) Limited
Regional Internet Registry	APNIC
Country	PK
Continent	AS

Registration Data (RDAP)

IP Version: v4

Address Range: 154.57.0.0 - 154.57.255.255



Selected hash SHA256: f4ec2dba070a9ecd1dee342329b0cd7caa406cd4995c41b9aab9aea9b73cd06d — was checked in VirusTotal.

5.2. Detection Tab

42

/ 59

Community Score

-2

42/59 security vendors flagged this file as malicious

Reanalyze Similar More

f4ec2dba070a9ecd1dee342329b0cd7caa406cd4995c41b9aab9aea9b73cd06d

f4ec2dba070a9ecd1dee342329b0cd7caa406cd4995c41b9aab9aea9b73cd06d.exe

Size445.50 KB

Last Analysis Date6 hours ago

EXE

peexe64bitsdetect-debug-environmentspreader

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY8

Crowdsourced IDS rules

HIGH 0MEDIUM 1LOW 3INFO 0

Matches rule PROTOCOL-ICMP Unusual PING detected at Snort registered user ruleset

successful-recon-limited

Matches rule PROTOCOL-ICMP PING Windows at Snort registered user ruleset

misc-activity

Matches rule PROTOCOL-ICMP PING at Snort registered user ruleset

misc-activity

Matches rule PROTOCOL-ICMP Echo Reply at Snort registered user ruleset

misc-activity

f4ec2dba070a9ecd1dee342329b0cd7caa406cd4995c41b9aab9aea9b73cd06d

Aruzhan Abdullina

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY8

Contacted URLs (3)

ScannedDetectionsStatusURL

2026-09-210/91200http://crl.globalsign.net/primobject.crl

2026-09-230/91404http://crl.globalsign.net/ObjectSign.crl

2026-09-080/91200http://crl.globalsign.net/Root.crl

Contacted Domains (6)

DomainDetectionsCreatedRegistrar

cdn.globalsigncdn.com.cdn....0/91- -

crl.globalsign.net0/911998-02-18 -

global.prn.cdn.globalsign.com0/911999-04-19 -

globalsign.net0/911998-02-18 -

res.public.onecdn.static.mic...0/912023-05-05 MarkMonitor Inc.

www.microsoft.com0/911991-05-02 -

Contacted IP addresses (10)

5.3. Details Tab

f4ec2dba070a9ecd1dee342329b0cd7caa406cd4995c41b9aab9aea9b73cd06d

Aruzhan Abdullina

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY8

Basic properties

MD53aa5fce520c9dc4a858c2317eb1b328

SHA-149f07b54c01fb1762cad915ee76301cc584f50

SHA-256f4ec2dba070a9ecd1dee342329b0cd7caa406cd4995c41b9aab9aea9b73cd06d

Vhash04506666d15551550f827932a3a3023211z

Authentihasha7c22660143d2b3c9b9f8b3ee667f08a9e2a30cebe93a2a290406a825c0086de

Imphash7ade23c90b842d6a9620c740656bcbfa

Rich PE header hash155a921041bd7a8dd6ef2758dde5784

SSDEEP6144:frcIz/RyJolcJHwHs4s8OmdpmfU6XQV3LOc2n6qxcEjxdCx+n36GdPmPDO+f9rcIz/Rzwlcl/t8SUTC6qiz8+nqQ6

TLSH11ABA46C96B6A618FDD066C07483478672BA7674850B31BDEF02E4C6312F26AE16F3DF50

File typeWin32 EXE executable windows win32 pe peexe

MagicPE32+ executable (GUI) x86-64, for MS Windows

TrIDWin64 Executable (generic) (33.1%) | Win16 NE executable (generic) (25.6%) | Windows Icons Library (generic) (10.4%) | OS/2 Executable (generic) (10.3%) | Generic Win/DOS Executa...

DetectItEasyPE64 Compiler: (x86_64-pc-windows-msvc) | Compiler: Microsoft Visual C/C++ (19.36.32533) [C] | Linker: Microsoft Linker (14.36.32824) | Tool: Visual Studio (2022 version 17.6)

MagikaPEBIN

File size445.50 KB (456192 bytes)

History

Creation Time2024-06-24 06:30:47 UTC

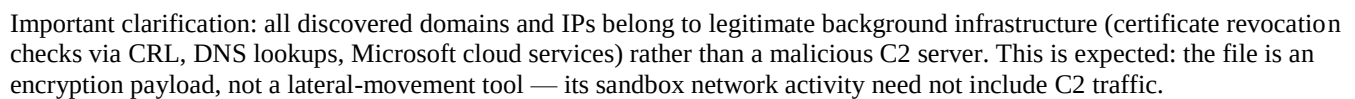
First Submission2024-07-10 17:29:46 UTC

Last Submission2026-09-23 10:40:47 UTC

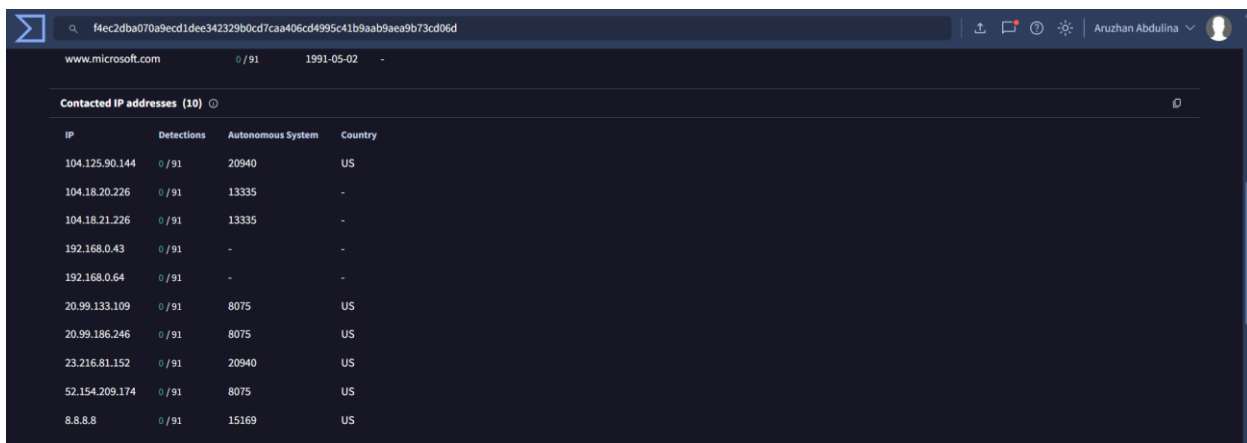
Last Analysis2026-09-24 05:49:43 UTC

Names

5.4. Relations Tab



5.5. Behavior Tab — Behavioral Analysis



www.microsoft.com 0 / 91 1991-05-02 -

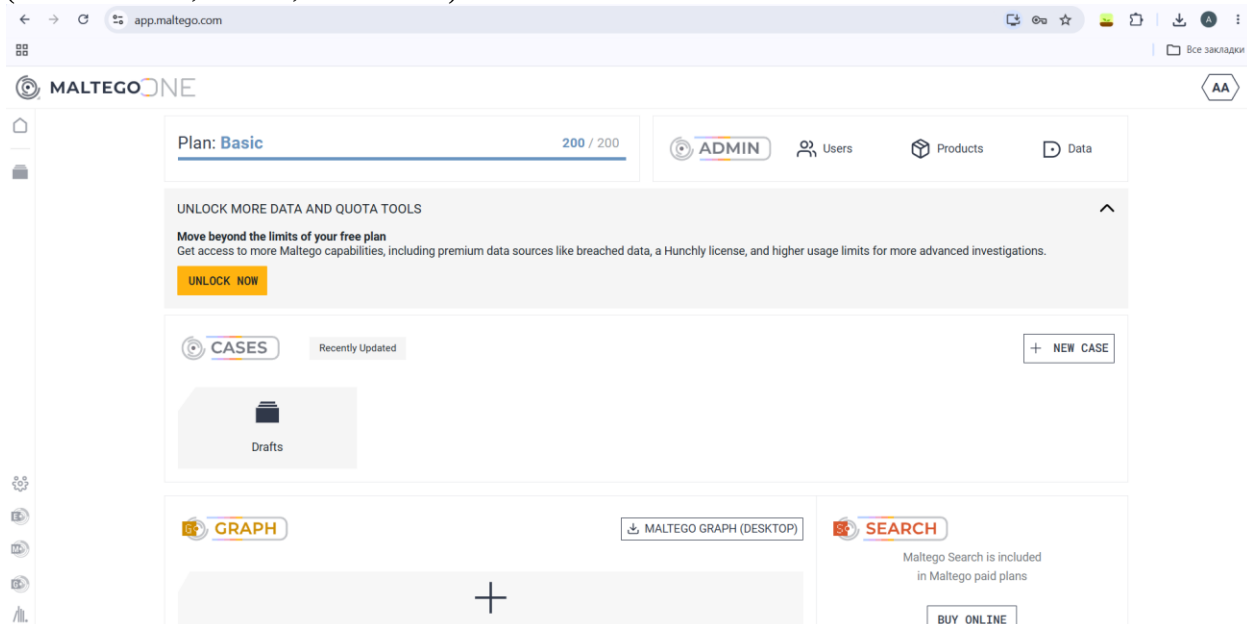
Contacted IP addresses (10)

IP	Detections	Autonomous System	Country
104.125.90.144	0 / 91	20940	US
104.18.20.226	0 / 91	13335	-
104.18.21.226	0 / 91	13335	-
192.168.0.43	0 / 91	-	-
192.168.0.64	0 / 91	-	-
20.99.133.109	0 / 91	8075	US
20.99.186.246	0 / 91	8075	US
23.216.81.152	0 / 91	20940	US
52.154.209.174	0 / 91	8075	US
8.8.8.8	0 / 91	15169	US

No direct match to T1021.006 was found in the sample's own behavior — this is expected and logically explainable: the analyzed sample is a ransomware payload deployed AFTER the lateral-movement stage, not the remote-access tool itself. According to the Microsoft Threat Intelligence report (Section 3), it is at the earlier stage of the attack that Storm-0501 uses Evil-WinRM (T1021.006) to gain access to the victim host, subsequently deploying payloads such as this one. The attack chain therefore logically links T1021.006 (initial access via WinRM) to the observed behavior: T1059 (command execution), T1027 (obfuscation), T1574 (persistence via hijack execution flow).

6. Maltego — Visualizing Infrastructure Relationships

The entry point for the graph was one of the IP addresses discovered in the VirusTotal Relations tab — 20.99.133.109 (Microsoft Azure, AS8075, United States).



app.maltego.com

MALTEGO ONE

Plan: Basic 200 / 200

ADMIN Users Products Data

UNLOCK MORE DATA AND QUOTA TOOLS

Move beyond the limits of your free plan
Get access to more Maltego capabilities, including premium data sources like breached data, a Hunchly license, and higher usage limits for more advanced investigations.

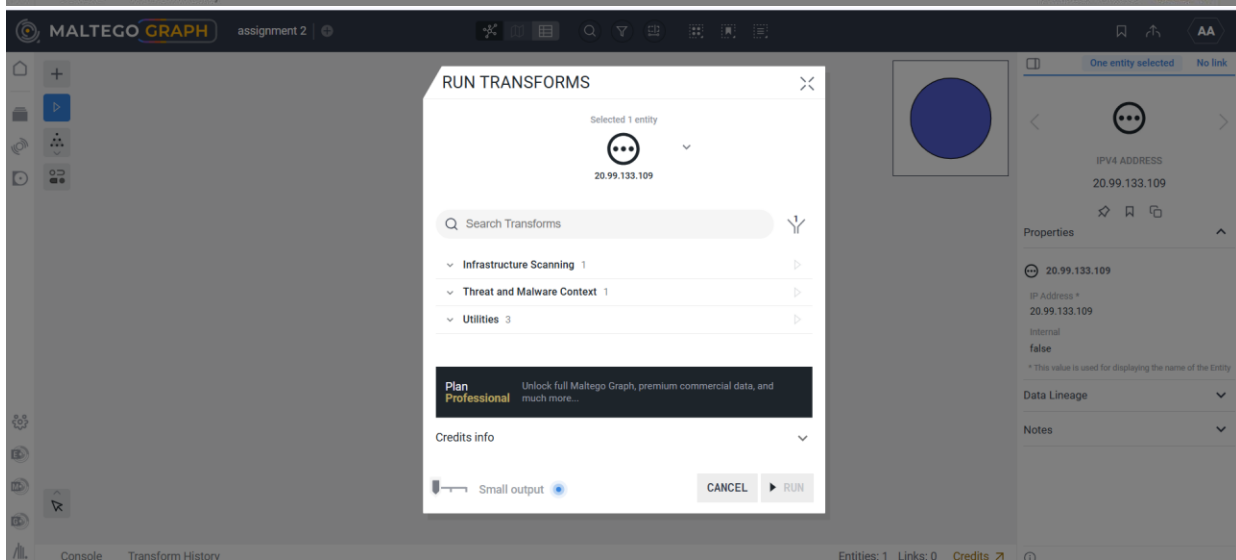
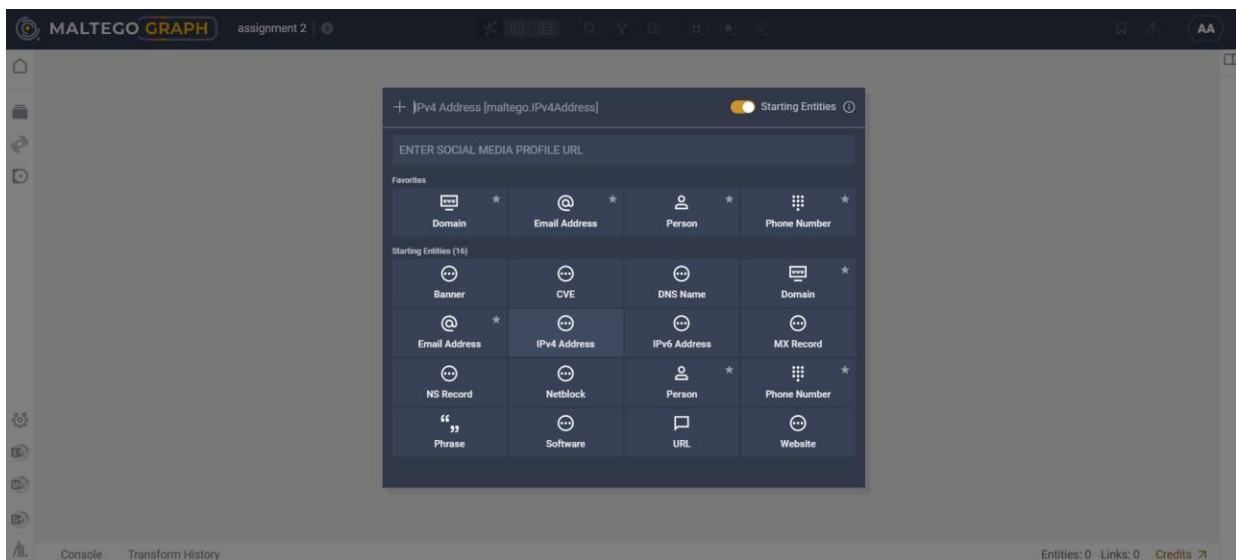
UNLOCK NOW

CASES Recently Updated + NEW CASE

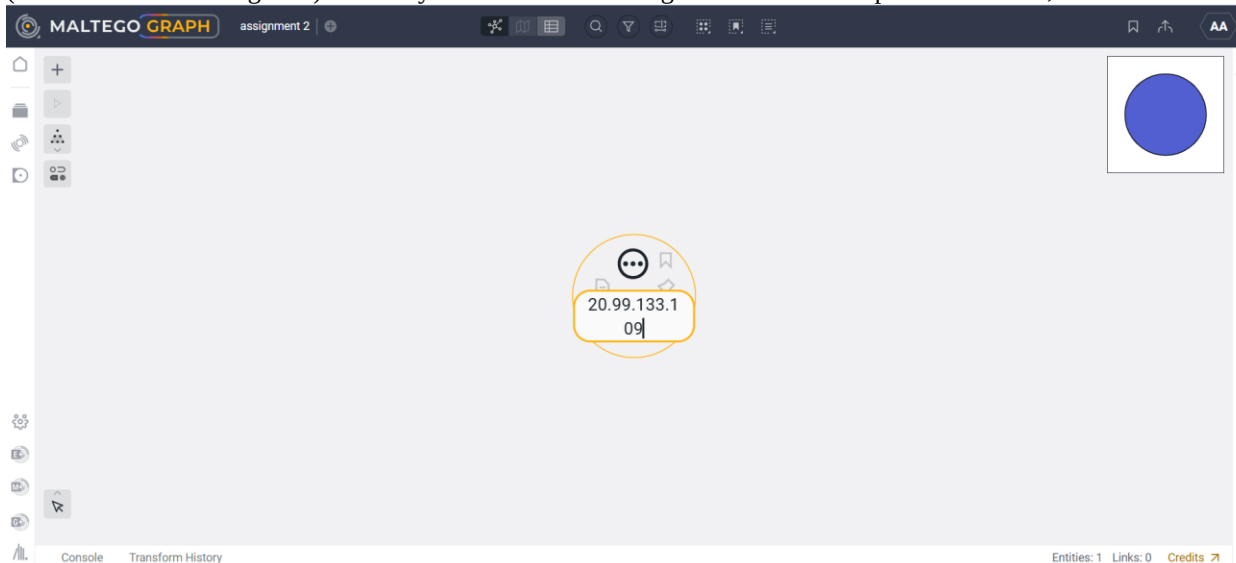
Drafts

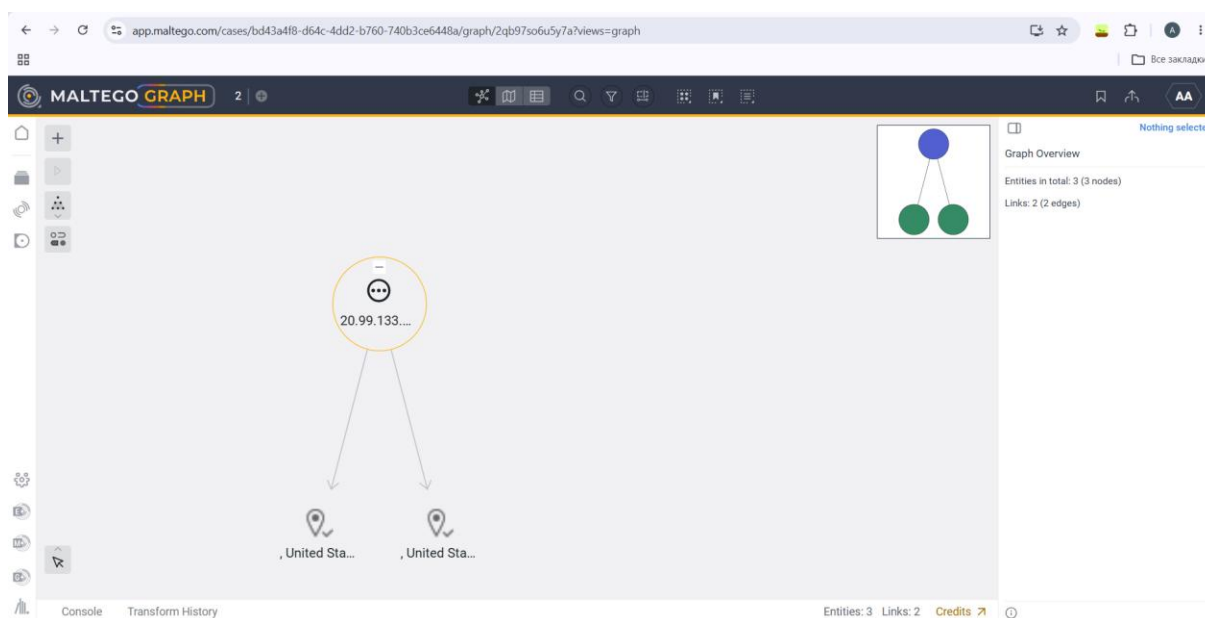
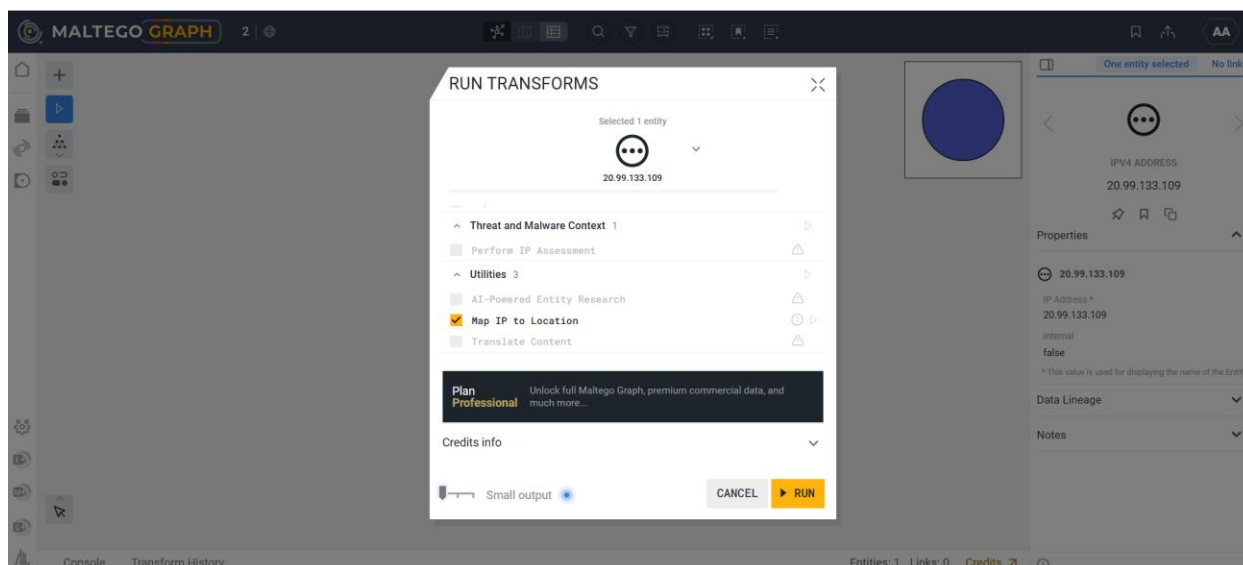
GRAPH MALTEGO GRAPH (DESKTOP)

SEARCH Maltego Search is included in Maltego paid plans BUY ONLINE



In the free Maltego Community Edition, most transforms are unavailable without a subscription to commercial data providers (marked with a warning icon). The only available and working transform was Map IP to Location, in the Utilities category.





The result confirms data previously obtained through VirusTotal (Autonomous System 8075 — Microsoft, country US): the geolocation independently matched across two different tools, which is a practical demonstration of Bazzell's source-triangulation methodology.

Research limitation: Maltego Community Edition provides a heavily restricted set of free transforms; building a full netblock/ASN/passive-DNS graph would require a paid plan (Maltego Professional).

7. Data Source Mapping

Summary table mapping the data sources used to investigate technique T1021.006:

Source	Type	What was obtained	Relevance to the theme (T1021.006)
MITRE ATT&CK	OSINT (authoritative)	Formal definition of technique T1021.006; list of groups/tools (Cobalt Strike, Brute Ratel C4, Chimera, Storm-0501/Evil-WinRM)	Frames the whole investigation and anchors every other source
Microsoft Threat Intelligence (Storm-0501 report)	OSINT (vendor report)	Description of a real incident: lateral movement via Evil-WinRM, DCSync, deployment of a ransomware payload	Direct documented confirmation of WinRM/PowerShell abuse in an active attack
Shodan	OSINT (scanner)	1,535,423 hosts with ports 5985/5986 open; example host 154.57.211.62 (Pakistan) with WinRM banner data (OS, NetBIOS domain)	Shows the real-world scale of exposed attack surface for T1021.006
MalwareBazaar (abuse.ch)	OSINT (IOC repository)	Tag-confirmed sample of the Embargo ransomware family, SHA256 f4ec2dba070a9ecd1dee342329b0cd7caa406cd4995c41b9aab9aea9b73cd06d	Provides a verified payload sample deployed after lateral movement via WinRM
VirusTotal	OSINT (reputation / sandbox)	42/59 detections; behavioral analysis (T1059, T1027, T1574); Contacted IPs/Domains/URLs	Confirms the sample's maliciousness and reveals post-exploitation behavior that complements the T1021.006 chain
Maltego	OSINT (aggregator)	Relationship graph: IP 20.99.133.109 → geolocation (US) via the Map IP to Location transform	Visualizes infrastructure context and independently confirms VirusTotal data (AS8075 / Microsoft Azure, US)

8. Conclusion

The Week 2 assignment implemented a full OSINT data-collection cycle around a single theme — Remote PowerShell attacks via WinRM (T1021.006):

- MITRE ATT&CK established the formal technical and contextual framework for the investigation, including a real case involving the group Storm-0501.
- Shodan revealed the real-world scale of exposed attack surface (1.5M+ exposed hosts) and demonstrated the information-disclosure risk through service banners.
- MalwareBazaar and VirusTotal made it possible to obtain and verify a real, confirmed ransomware payload sample (Embargo), linked to the investigated technique via the Microsoft report.
- Maltego visualized the infrastructure context and independently confirmed the data obtained through VirusTotal.

The main methodological takeaway, consistent with Bazzell's approach: no single source provides a complete picture — only the consistent cross-referencing of several independent tools (triangulation) allows building a reliable, verifiable analysis chain, from a formal technique (MITRE) to a concrete malicious sample (VirusTotal) and its associated infrastructure (Shodan, Maltego).